

PLC User Manual

Chengdu CRP Robot Technology Co., Ltd

Website: www.crprobot.com Tel: 028-84203568 Fax: 028-83951758

Add: No.42, Huatai Road, Chenghua District, Chengdu City, China.

Contents

1. Overview	3
2. Basic component symbols in ladder diagram	3
3. Basic logic commands in command list	4
3.1 LD, LDI, OUT command	4
3.2 AND and ANI command	4
3.3 OR and ORI command	4
3.4 ORB command	4
3.5 ANB command	4
3.6 MPS, MRD, MPP stack memory and multiple output command	4
3.7 SET and RST command	4
3.8 NOP and END command	5
4. Component description	5
4.1 Component list	5
4.2 Detailed description of components	5
4.2.1 Input relay (X)	5
4.2.2 Output relay (Y)	6
4.2.3 Internal relay (M)	6
4.2.4 Timer (T)	6
4.2.5 Counter (C)	7
5. PLC resources of CD60-CRX8 system	8
5.1 The distribution of the input and output interfaces in the main box	9
5.1.1 Robot dedicated terminal	9
5.1.2 Common output interface	10
5.1.3 Common input interface	11
5.1.4 Teach pendant input and output interface	12
5.1.5 Remote box interface distribution	13
5.1.6 Enable signal	13
5.2 Definition of system dedicated auxiliary relay	13
6. Editing of PLC program	39
6.1 PLC ladder diagram examples	39
7. PLC program backup and update	40
7.1 PLC backup	40
7.2 PLC update and replacement	42

1. Overview

The control system has a software PLC function. According to the system's programmable I/O, auxiliary relays, timers and other resource management system all I/O interfaces and internal auxiliary relays, logic judgment, counting and other processing can be performed to realize the detection of input ports and the control of output ports as well as and The system kernel performs data exchange.

For different systems, the PLC functional indicators are different. The PLC indicators of the CD60-CRX8 system are as follows:

CD60-CRX8 PLC indicators	
Edit mode	Ladder diagram
Execute commands	Command list
Program capacity	10000 steps
Execution cycle	10ms
Execution way	①Sequential execution, ②Circular execution
Basic commands	Internal relay, timer, counter, input relay X, output relay Y

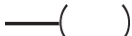
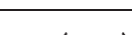
Note:

①Sequential execution: the execution of PLC is to execute step by step the program file of the command list line by line, which is not exactly the same as the working principle of the general relay circuit.

②Cyclic execution: PLC executes from the beginning of the ladder diagram (command list) to the end of the ladder diagram. After the ladder diagram (command list) ends, the execution restarts from the beginning of the ladder diagram (command list) again.

2. Basic component symbols in ladder diagram

In order to facilitate the editing of the ladder diagram in the CD60-CRX8 system, the following basic component symbols are specially set:

No.	Symbolic name	Symbolic graphics	Description
1	Normally open contact		Normally open contacts of each element (relay)
2	Normally closed contact		Normally closed contacts of each element (relay)
3	Direct output coil		Coils of components (relays)
4	Set output coil		Coil of each element (relay) (set hold)
5	Reset output coil		Coil of each element (relay) (reset hold)

6	Horizontal connection line		Used to connect the contacts of each relay horizontally and coil
7	Vertical connection line		Used to connect the contacts and coils of each relay vertically (usually for branching)

3. Basic logic commands in command list

In the CD60-CRX8 system, there are 16 basic logic commands in the PLC command list, which are used to achieve basic logic control.

3.1 LD, LDI, OUT command

LD: Normally open contact and bus connection command.

LDI: Normally closed contact and bus connection command .

OUT: Relay coil output command.

LD and LDI commands can be used for X, Y, M, T and C. They can also be used with AND and ORB commands for the starting point of a branch circuit. OUT command can be used for Y, M, T and C, but cannot be used for input relay X.

3.2 AND and ANI command

AND: Normally open contact series connection command.

ANI: Normally closed contact series connection command.

AND and ANI commands can be used for X, Y, M, T, and C.

3.3 OR and ORI command

OR: Normally open contact parallel connection command.

ORI: Normally closed contact parallel connection command.

OR and ORI commands can be used for X, Y, M, T and C.

3.4 ORB command

ORB: Parallel connection command for series circuit blocks.

A circuit formed by connecting two or more contacts in series is called a "series circuit block". When connecting a series circuit block in parallel, use the ORB command.

3.5 ANB command

ANB: Series connection command for parallel circuit blocks.

The ANB command connects the parallel circuit block in series with the previous circuit. Before using the ANB command, the internal connection of the parallel circuit block should be completed.

3.6 MPS, MRD, MPP stack memory and multiple output command

The MPS, MRD, and MPP commands are the in-stack, read-stack, and out-stack commands, respectively, and are used for multiple output circuits.

3.7 SET and RST command

Set command: an command to hold an operation.

Reset command: an command to keep the operation reset.

SET command can be used for Y and M, RST command can be used for Y, M, T and C.

3.8 NOP and END command

NOP: null operation command.

END: end command, which means the end of the program.

4. Component description

The component is the address symbol used to identify the signal processed in the PLC. The component number is the serial number assigned to the component. The component numbers of components X, Y and M in the PLC of this system are expressed in decimal notation.

4.1 Component list

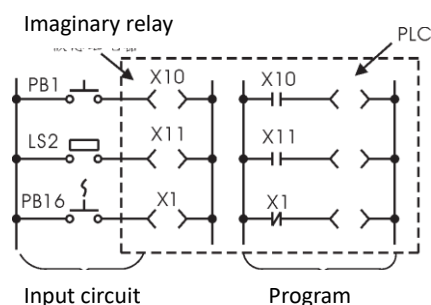
The following table is a list of all components of the PLC system.

Component list	Description
X	System input signal
Y	System output signal
T	0ms general timer; 10ms totalizer timer; 100ms general timer; 100ms totalizer timer; 1s general timer; 1s totalizer timer.
C	System counter
M	System auxiliary relay

4.2 Detailed description of components

4.2.1 Input relay (X)

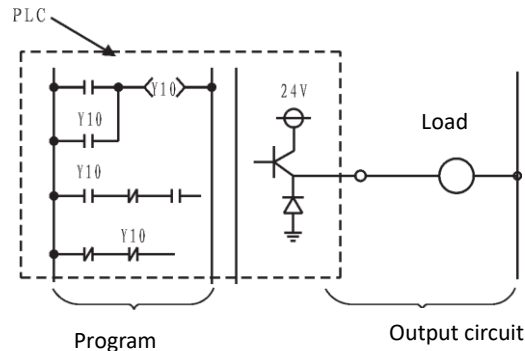
The input relay is the window through which the system PLC receives the external input switch signal. Commands from external components such as buttons, changeover switches, limit switches or digital switches are introduced into the system by the input point circuit; Assuming that each input point corresponds to a relay Xn, the PLC program can use the normally open and normally closed contacts of the relay for program editing, and can be used multiple times. As shown below:



Note: The input relay number is expressed via decimal value.

4.2.2 Output relay (Y)

The output relay system PLC sends the signal window to the external load. The output relay is used to transmit the output signal of the system PLC to the output point, which then drives the external load. Each relay of the output point has only one pair of normally open points, but the normally open contacts and normally closed contacts of each output relay in the ladder diagram can be used (and can be used multiple times), as shown in the following figure.



Note: The output relay number is expressed via decimal value.

4.2.3 Internal relay (M)

The internal relay is an auxiliary relay in the PLC and cannot be directly output to an external source. It is usually defined as the input, output or status of a certain function, and is the channel for information transmission between the PLC and the core control software of the system. He has the following characteristics:

- (a) These relays are cleared when the power is turned off.
- (b) Normally open contacts and normally closed contacts of internal relays that can be used in ladder diagrams
- (c) The internal relay number is expressed in decimal numbers. Most internal relays have been defined by the system for a certain function and control mode, and only some internal relays are used as temporary intermediate signals.

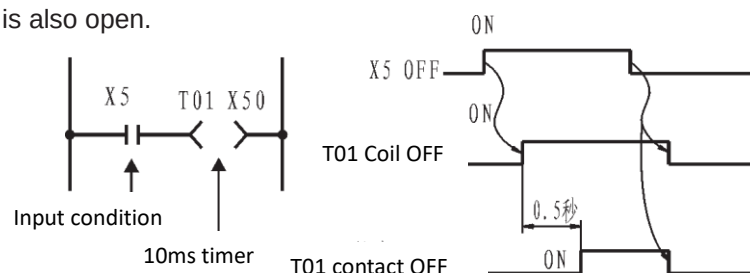
4.2.4 Timer (T)

The timer in the system PLC is equivalent to the time relay in the relay system. In this system, the timer is divided into general timer and totalizer timer. The constant K is used as the set value of the timer.

General timer

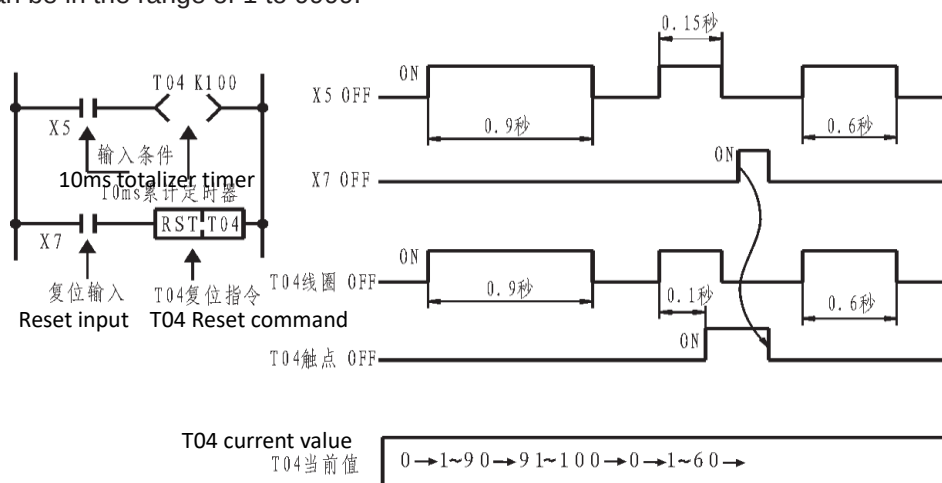
Its working principle is: when the input condition is met, start counting, when the timing reaches the set value, the timer contact is turned on; if the input condition is turned off, the timer count value will be set to 0, and the contact will also be turned off open. The timer setting value (K) is a decimal number, which can be in the range of 1 to 9999.

Example: As shown in Figure 5.3, when the input relay X5 is valid, the timer T01 counts when it is on. After 0.5 seconds (K value is 50, T01 is a 10ms timer), the T01 contact turns on. When the input relay X5 is invalid, the timer T01 If it is set to 0, the contact is also open.



Totalizer timer

Its working principle is: when the input condition is met, start counting, when the timing reaches the set value, the timer contact is turned on; even if the input condition is turned off, the current value (counting value) of the accumulative timer will remain and the contact state Does not change. When the RST command is executed, the count value of the accumulative timer will be set to 0, and the contact will be disconnected. The timer setting value (K) is a decimal number, which can be in the range of 1 to 9999.



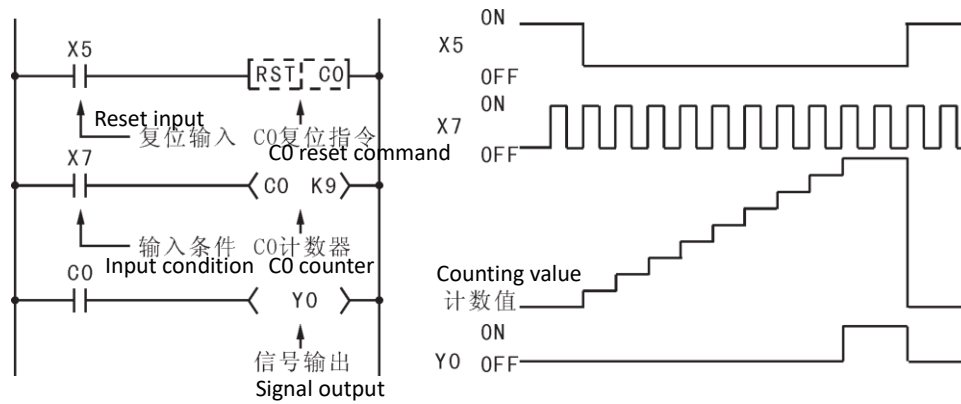
Example: As shown in Figure 5.4, when the input relay X5 is valid, the timer T04 counts when it is on. After 1 second (K value is 100, T04 is a 10ms timer), the T04 contact is on, as shown in the waveform diagram, even if the input condition X5 is off. On, the current value (counting value) of the accumulative timer T04 will still be maintained, and the contact status will not change. When the input relay X7 is valid, the timer T04 is reset to 0, the contact is also disconnected, and when the input signal is valid again, T04 starts counting again.

4.2.5 Counter (C)

The counter is used to accumulate and detect the rising edge of the input condition (X, Y, M, T). Therefore, when the input condition is ON, the count does not change. The response cycle is 100ms. Therefore, the on and off duration of the input condition should be greater than 100ms. The PLC of this system has 16 counters C01-C15, which have the following characteristics:

- The count value is a decimal number and can be specified in the range of 1 to 9999.
- Even if the count value of the counter is not cleared when the input condition is closed, the count value of the counter must be cleared by the RST command
- The normally open contacts and normally closed contacts of the counter can be used in the ladder diagram.

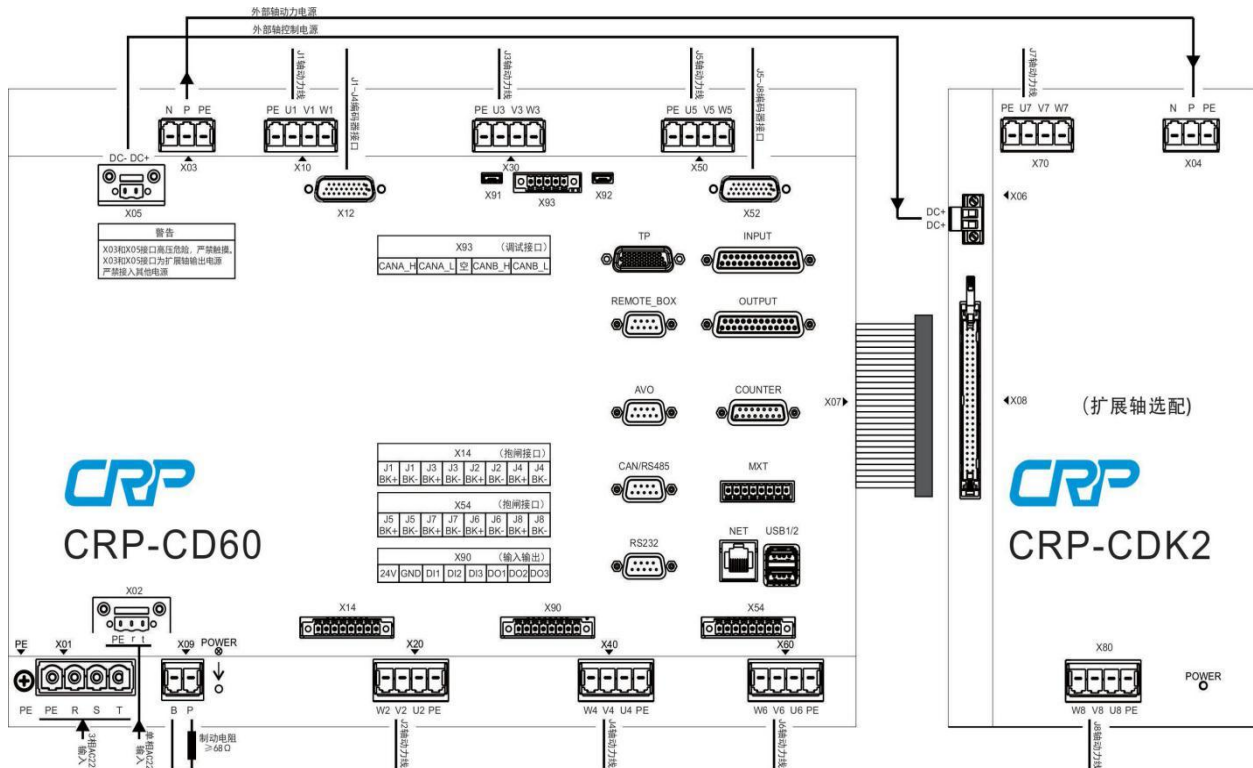
Example: As shown in the figure below, after the X5 input impedance signal is valid, C0 is reset, its corresponding count value is set to "0", and its normally open contact is disconnected (the normally closed contact is connected). The X7 input signal is used to provide the counting input signal. When the reset input of the counter is disconnected and the counting input signal changes from disconnected to connected (that is, the rising edge of the counting pulse), the current value of the counter is increased by 1. After 9 input count pulses, the C0 counter is valid (constant K is 9), its normally open contact is on (the normally closed contact is off), and the Y0 output is valid. If the pulse counter count value remains unchanged until the reset input signal is valid, the count value of the counter is set to "0".



5. PLC resources of CD60-CRX8 system

Component	Number	Quantity	Description
X	X00-X22	23	universal input interface
	X24-X46	23	In reserve
	X48-X63	16	Motion control input interface
	X64-X88	25	Robot teach device and dedicated terminal input interface
Y	Y00-Y23	24	Universal output interface
	Y24-Y46	22	In reserve
	Y48-Y55	8	Motion control output interface
	Y56-Y63	8	Robot teach device and dedicated terminal input interface
T	T00-T09	10	10ms general-purpose timer
	T10-T19	10	10ms totalizer timer
	T20-T29	10	10ms general-purpose timer
	T30-T39	10	10ms totalizer timer
	T40-T49	10	1s general timer
	T50-T59	10	1s totalizer timer
C	C00-C19	20	System counter
M	M00-M499	500	System dedicated auxiliary relay
	M500-M799	300	General auxiliary relay

5.1 The distribution of the input and output interfaces in the main box



5.1.1 Robot dedicated terminal

Function: connect the robot dedicated signal MXT			
Pin	Name	Definition	Description
1	X75	AF_SER	Anti-collision sensor
2	X76	EX_STOP1	External emergency stop 1
3	X77	EX_STOP2	External emergency stop 2
4	X78	SA_B01	Safety pin1
5	X79	SA_B02	Safety pin2
6	X81	X81 STOP_T	In reserve(do not connect)
7	-	Ground wire 0V	-
8	-	+24V	-

5.1.2 Common output interface

Function: I/O interface Y00 to Y23 output OUTPUT			
Pin	Name	Definition	Description
1	Y00	Common output port	-
2	Y02	Common output port	-
3	Y04	Common output port	-
4	Y06	Common output port	-
5	Y08	Common output port	-
6	Y10	Common output port	-
7	Y12	Common output port	-
8	Y14	Common output port	-
9	Y16	Common output port	-
10	Y18	Common output port	-
11	Y20	Common output port	-
12	Y22	Common output port	-
13	(output) +24V	+24V (power input)	-
14	Y01	Common output port	-
15	Y03	Common output port	-
16	Y05	Common output port	-
17	Y07	Common output port	-
18	Y09	Common output port	-
19	Y11	Common output port	-
20	Y13	Common output port	-
21	Y15	Common output port	-
22	Y17	Common output port	-
23	Y19	Common output port	-
24	Y21	Common output port	-
25	output +24V	+24V (power input)	-

5.1.3 Common input interface

Function: I/O interface X00 to X22 input			
Pin	Name	Definition	Description
1	X00	Common output port	-
2	X02	Common output port	-
3	X04	Common output port	-
4	X06	Common output port	-
5	X08	Common output port	-
6	X10	Common output port	-
7	X12	Common output port	-
8	X14	Common output port	-
9	X16	Common output port	-
10	X18	Common output port	-
11	X20	Common output port	-
12	X22	Common output port	-
13	GND	Ground wire 0V (power input)	-
14	X01	Common output port	-
15	X03	Common output port	-
16	X05	Common output port	-
17	X07	Common output port	-
18	X09	Common output port	-
19	X11	Common output port	-
20	X13	Common output port	-
21	X15	Common output port	-
22	X17	Common output port	-
23	X19	Common output port	-
24	X21	Common output port	-
25	GND	Ground wire 0V (power input)	-

5.1.4 Teach pendant input and output interface

Function: Connect teaching box TP			
Pin	Name	Definition	Description
1-5	-	-	-
6	X66	E_STOP	Emergency stop signal
7	X67	SWITCH1	Select switch 1
8	-	GND	-
9	-	+24V	-
10-15	-	-	-
16	X70	SAFE-SW2	Safety switch 2
17	Y59	LCD_CON	LCD lamp control
18	-	+24V	-
19	-	-	-
20	-	-	-
21	X71	SAFE-SW3	Safety switch 3
22-23	-	-	-
24	X69	SAFE-SW1	Safety switch 1
25	X68	SWITCH2	Select switch 2
26	-	GND	-

5.1.5 Remote box interface distribution

Function: Connect to the remote operation box of the robot Remote Box			
Pin	Name	Definition	Description
1	X84	EX_RUN	Remote operation
2	X85	EX_HALT	Remote stop
3	X86	EX_POWER	Externally enable
4	X87	EX_RESET	Remote reset
5	Y60	RUN_OUT	Running
6	Y61	SERVO_ON	Servo is on
7	Y62	ALM_OUT	Alarm Output
8	-	GND	-
9	-	+24V	-

5.1.6 Enable signal

Name	Definition	Description
Y49	SRV-ON	Enable

5.2 Definition of system dedicated auxiliary relay

No.	Code	Name	Character	Description
1	M00	Emergency stop	state	When the auxiliary relay is valid, the system is in emergency stop state, and its contacts can also be used for control.
2	M01	Just started	state	The auxiliary relay can start the robot as soon as the system is started, indicating that it has just been turned on.
3	M02	Teach mode	state	The auxiliary relay can start the robot when the system is in teaching mode.
4	M03	Play mode	state	The auxiliary relay can start the robot when the system is in the reproduction mode.
5	M04	Remote mode	state	The auxiliary relay can start the robot when the system is in remote mode.
6	M05	Button start mode	state	The auxiliary relay is valid when the system is in button

				start mode.
7	M06	Handwheel start mode	state	When the system is in handwheel start mode, the auxiliary relay can start the robot.
8	M07	Forbidding starting mode	state	The auxiliary relay can start the robot when the system is in forbidding starting mode.
9	M08	Safety switch	state	When the safety switch is pressed, the auxiliary relay can start the robot.
10	M09	Allowed to start	input	When the safety switch is valid, the auxiliary relay is valid, and the robot can be up and running. It can also run without an alarm.
11	M10	Decelerate to stop	input	
12	M11	Fast stop	input	
13	M12	Run	input	In automatic operation, the auxiliary relay activates the robot and the robot runs.
14	M13	Anti-collision sensor	input	When the auxiliary relay activates the robot, it means the anti-collision sensor works
15	M14	Maintenance mode	input	When the auxiliary relay activates robot, it means the safety pin signal action
16	M15	Full speed operation mode	input/state	When the auxiliary relay activates the robot, it means that the signal moves at full speed
17	M16	Brake completion	input	When the auxiliary relay activates robot, it means that the brakes of the robot are all released.
18	M17	Servo ready	input	Each axis drive is ready
19	M18	Absolute encoder read in state	state	Read absolute code wheel position
20	M19	Alarm control	output	Used to control the brake output
21	M20	Servo enable	output/state	When the auxiliary relay is valid, it means that the servo motor has been powered on
22	M21	Running	output /state	When the auxiliary relay is valid, it means that the robot and the external axis are in motion (light on the running light)
23	M22	In stopping	output /state	When the auxiliary relay is valid, it means that the robot and the external axis are stopping (light on the stop light)
24	M23	The robot is abnormal	output /state	When the auxiliary relay is valid, it means that the robot and external axis are in an abnormal state such as an alarm
25	M24	Alarm (abnormal) reset (R)	output	When the system presses the "reset" button in the special interface when the emergency stop or the robot is abnormal, the status relay is valid for 2 seconds and then cancelled

26	M25	Emergency stop mode	input/state	When the auxiliary relay is valid, it means that the robot is in the hardware emergency stop state
27	M26	Power on		Green icon
28	M27	Pulse sending completed		
29	M28	Remote operation	input	To run the program when this relay is active, use the external run button.
30	M29	Remote stop	input	Stop the program when this relay is active, use an external stop button.
31	M30	J1 axis drive alarm	input	When the auxiliary relay is valid, it indicates the J1 axis drive alarm
32	M31	J1 axis drive ready	input	When the auxiliary relay is valid, it means that the J1 axis drive is ready
33	M32	J2 axis drive alarm	input	When the auxiliary relay is valid, it means J2 axis drive alarm
34	M33	J2 axis drive ready	input	When the auxiliary relay is valid, it means that the J2 axis drive is ready
35	M34	J3 axis drive alarm	input	When the auxiliary relay is valid, it indicates the J3 axis drive alarm
36	M35	J3 axis drive ready	input	When the auxiliary relay is valid, it means that the J3 axis drive is ready
37	M36	J4 axis drive alarm	input	When the auxiliary relay is valid, it indicates the J4 axis drive alarm
38	M37	J4 axis drive ready	input	When the auxiliary relay is valid, it means that the J4 axis drive is ready
39	M38	J5 axis drive alarm	input	When the auxiliary relay is valid, it means J5 axis drive alarm
40	M39	J5 axis drive ready	input	When the auxiliary relay is valid, it means that the J5 axis drive is ready
41	M40	J6 axis drive alarm	input	When the auxiliary relay is active, it indicates the J6 axis drive alarm
42	M41	J6 axis drive ready	input	When the auxiliary relay is active, it means that the J6 axis drive is ready.
43	M42	J7 axis drive alarm	input	When the auxiliary relay is valid, it indicates the J7 axis drive alarm
44	M43	J7 axis drive ready	input	When the auxiliary relay is valid, it indicates that the J7 axis drive is ready
45	M44	J8 axis drive alarm	input	When the auxiliary relay is valid, it indicates the J8 axis drive alarm

46	M45	J8 axis drive ready	input	When the auxiliary relay is valid, it means that the J8 axis drive is ready
47	M46	the position of J1 axis is out of tolerance	state	When the position of J1 axis is out of tolerance, it is valid.
48	M47	the position of J2 axis is out of tolerance	state	When the position of J2 axis is out of tolerance, it is valid.
49	M48	the position of J3 axis is out of tolerance	state	When the position of J3 axis is out of tolerance, it is valid.
50	M49	the position of J4 axis is out of tolerance	state	When the position of J4 axis is out of tolerance, it is valid.
51	M50	the position of J5 axis is out of tolerance	state	When the position of J5 axis is out of tolerance, it is valid.
52	M51	the position of J6 axis is out of tolerance	state	When the position of J6 axis is out of tolerance, it is valid.
53	M52	the position of J7 axis is out of tolerance	state	When the position of J7 axis is out of tolerance, it is valid.
54	M53	the position of J8 axis is out of tolerance	state	When the position of J8 axis is out of tolerance, it is valid.
55	M54	J1 axis returns to zero point, deceleration switch of J1 axis is valid	input	
56	M55	J2 axis returns to zero point, deceleration switch of J1 axis is valid	input	
57	M56	J3 axis returns to zero point, deceleration switch of J1 axis is valid	input	
58	M57	J4 axis returns to zero point, deceleration switch of J1 axis is valid	input	
59	M58	J5 axis returns to zero point, deceleration switch of J1 axis is valid	input	

60	M59	J6 axis returns to zero point, deceleration switch of J1 axis is valid	input	
61	M60	J7 axis returns to zero point, deceleration switch of J1 axis is valid	input	
62	M61	J8 axis returns to zero point, deceleration switch of J1 axis is valid	input	
63	M62	The positive hardware limit of J1 is valid	input	
64	M63	The positive hardware limit of J2 is valid	input	
65	M64	The positive hardware limit of J3 is valid	input	
66	M65	The positive hardware limit of J4 is valid	input	
67	M66	The positive hardware limit of J5 is valid	input	
68	M67	The positive hardware limit of J6 is valid	input	
69	M68	The positive hardware limit of J7 is valid	input	
70	M69	The positive hardware limit of J8 is valid	input	
71	M70	The negative hardware limit of J1 is valid	input	
72	M71	The negative hardware limit of J2 is valid	input	
73	M72	The negative hardware limit of J3 is valid	input	

74	M73	The negative hardware limit of J4 is valid	input	
75	M74	The negative hardware limit of J5 is valid	input	
76	M75	The negative hardware limit of J6 is valid	input	
77	M76	The negative hardware limit of J7 is valid	input	
78	M77	The negative hardware limit of J8 is valid	input	
79	M78	J1 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
80	M79	J2 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
81	M80	J3 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
82	M81	J4 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
83	M82	J5 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
84	M83	J6 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
85	M84	J7 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
86	M85	J8 axis return to zero state relay	state	The relay is valid, indicating that the return to zero point has been completed
87	M86	J1 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
88	M87	J2 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
89	M88	J3 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
90	M89	J4 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
91	M90	J5 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed

92	M91	J6 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
93	M92	J7 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
94	M93	J8 axis servo enable complete	state	When the relay is valid, it means that the servo enable has been completed
95	M94		state	
96	M95		state	
97	M96		state	
98	M97		state	
99	M98		state	
100	M99		state	
101	M100	In play and remote mode, the signal is from valid to invalid, and return to the first line of the main program.	state	
102	M101	Abnormal strong current, fail to end the enable.	state	
103	M102	driver alarm	state	
104	M103	working original point 1	state	GP96
105	M104	working original point 2	state	GP97
106	M105	working original point 3	state	GP98
107	M106	working original point 4	state	GP99
108	M107	Maxsine Absolute Servo SEN	state	
109	M108	Power failure detection relay	state	Valid means that the strong power of the robot system is disconnected. Invalid means that the strong power of the system is normal
110	M109	Remote servo power-on button, press once to power on, then press again power off	input	

111	M110	Remote reset. If alarm, it is valid at remote mode	input	
112	M111	Prohibit to open the spray torch, 0--allow to open the torch	input	Y0
113	M112	Prohibit spray torch 1, 0--allow to open the torch	input	
114	M113	Prohibit spray torch 2, 0--allow to open the torch	input	
115	M114	Spray torch 1 gas open	output	Y1
116	M115	Spray torch 2 gas open	output	Y2
117	M116	Spray torch 1 open	output	Y3
118	M117	Spray torch 2 open	output	Y4
119	M118	Press to set to high, delay 100 ms to set to low, automatic cycle to complete one output, rotary disk	input	
120	M119	automatic program end and set to low	state	
121	M120	the forward signal of auto turntable (inching)	input	
122	M121	the reverse signal of auto turntable (inching)	input	
123	M122	the forward signal of auto turntable (continuous)	input	
124	M123	the reverse signal of auto turntable (continuous)	input	
125	M124	Valid, J7 axis exists on the J8 axis, and the two are the same	input	The pulse of J7 axis is copied to J8 axis, at this time, the pulses of J7 and J8 axis are same.

126	M125	Valid, J8 axis exists on the J7 axis, and the two are the same	input	The pulse of J8 axis is copied to J7 axis, at this time, the pulses of J7 and J8 axis are same.
127	M126	Valid, J7 axis runs while J8 axis stops	input	Prohibit J8 axis, J7 axis is normal
128	M127	Valid, J8 axis runs while J7 axis stops	input	Prohibit J7 axis, J8 axis is normal
129	M128	system alarm stop, robot set to 1, reset after 0	state	
130	M129			
131	M130	program running set to 1, stop set to 0	state	
132	M131	how much distance has passed point A in tracking, and when the next tracking begins, it will cancel	state	
133	M132			
134	M133			
135	M134			
136	M135	M135 is valid, record feedback joints		
137	M136	M136 rising edge, open the FEEDJ file, falling edge, close the FEEDJ file.		
138	M137			
139	M138			
140	M139	When alarm, clear zero point(1-do not clear zero point when alarm, 0-clear zero point when alarm)	input	
141	M140	Returning to zero point(output)	output	
142	M141	Return to zero point	output	

		is complete (output)		
143	M142	Fail to return to zero point (output)	output	
144	M143	Start automatically return to zero point (input)	output	
145	M144	Running to zero point	output	
146	M145			
147	M146			
148	M147			
149	M148	(CRX8) Emergency stop first, delay to turn off strong current	Input output	Emergency stop first, delay to turn off strong current
150	M149	(CRX8) energy saving tips	input	The robot enters energy-saving mode and the servo is powered off
151	M150	Already used		System PLC has been used
152	M151	Already used		System PLC has been used
153	M152	Already used		System PLC has been used
154	M153	Already used		System PLC has been used
155	M154	Already		System PLC has been used
156	M155	Already		System PLC has been used
157	M156	Already		System PLC has been used
158	M157	Already		After a certain time, if the servo enable is still not completed, it will be valid
159	M158	Already		Servo enable is completed
160	M159	Already	state	
161	M160	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
162	M161	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
163	M162	Manual m control of	output	M shortcut key, can be controlled by keyboard

		handling interface		
164	M163	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
165	M164	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
166	M165	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
167	M166	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
168	M167	Manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
169	M168	manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
170	M169	manual m control of handling interface	output	M shortcut key, can be controlled by keyboard
171	M170			
172	M171	0 -- normally closed 1-normally open, J1 axis servo alarm	state	The relay state can be set by servo parameters
173	M172	0 -- normally closed 1-normally open, J2 axis servo alarm	state	The relay state can be set by servo parameters
174	M173	0 -- normally closed 1-normally open, J3 axis servo alarm	state	The relay state can be set by servo parameters
175	M174	0 -- normally closed 1-normally open, J4 axis servo alarm	state	The relay state can be set by servo parameters
176	M175	0 -- normally closed 1-normally open, J5 axis servo alarm	state	The relay state can be set by servo parameters
177	M176	0 -- normally closed 1-normally open, J6 axis servo alarm	state	The relay state can be set by servo parameters
178	M177	0 -- normally closed 1-normally open, J7 axis servo alarm	state	The relay state can be set by servo parameters
179	M178	0 -- normally closed 1-normally open, J8 axis servo alarm	state	The relay state can be set by servo parameters

180	M179			
181	M180	Arc start detection signal	input	
182	M181	Abnormal detection of welding power source	input	
183	M182	Abnormal detection of water cooling	input	
184	M183	Point finding signal detection	input	
185	M184	Position finding start signal	output	
186	M185			
187	M186	Communication welding machine, valid automatically, feed wire	input	
188	M187	Communication welding machine, valid automatically, withdraw wire	input	
189	M188	Wire withdraw output signal, valid with M191 together	output	
190	M189	Air supply output signal	output	
191	M190	Arc start output signal	output	Analog output 1-current ; Analog output 2-voltage
192	M191	Wire feeding output signal	output	
193	M192	Point finding output signal	output	
194	M193	Whether the anti-collision sensor is valid	input	0-valid 1-invalid
195	M194			
196	M195		input	From invalid to valid, the speed becomes half of the current. If from valid to invalid, speed will recover.
197			input	M195 is valid, if it is from invalid to valid, means to stop. If from valid to invalid, means to run. If M195 is valid, the

	M196			speed becomes half, if M195 is invalid, the speed will recover.
198	M197		input	In the automatic remote state, running MOVL, MOVC is valid. Other indicators is invalid. If stop,it is invalid.
199	M198			
200	M199			
201	M200	Abnormal welding tongs	input	
202	M201	Pressurized to reach	input	
203	M202	Welding completed (standby)	input	
204	M203			
205	M204	Welding tongs alarm reset	output	
206	M205	Welding tongs reset	output	
207	M206	Welding starts	output	
208	M207	Position torque switching	output	
209	M208	Use IO signal to control the spot welder	input	
210	M209	Valid simulating welding	output	
211	M210	Focus + (click high level for 100 ms once)	input	
212	M211	Focus - (click high level for 100 ms once)	input	
213	M212	Effective range finder on	input	
214	M213	Effective camera on	input	
215	M214	Valid means forearm, invalid means posterior arm		
216	M215	The AS command controls the motor commutation and	output	

		after stopped, set to 0		
217	M216	Added M216 as an external IO to change. Continuous cycle and single cycle	input	
218	M217	Reverse to M215, set to 0 when stopped	output	
219	M218			
220	M219			
221	M220	Valid, forbid the alarm of out of tolerance detection of J1 axis	input	
222	M221	Valid, forbid the alarm of out of tolerance detection of J2 axis	input	
223	M222	Valid, forbid the alarm of out of tolerance detection of J3 axis	input	
224	M223	Valid, forbid the alarm of out of tolerance detection of J4 axis	input	
225	M224	Valid, forbid the alarm of out of tolerance detection of J5 axis	input	
226	M225	Valid, forbid the alarm of out of tolerance detection of J6 axis	input	
227	M226	Valid, forbid the alarm of out of tolerance detection of J7 axis	input	
228	M227	Valid, forbid the alarm of out of tolerance detection of J8 axis	input	
229	M228			

230	M229			
231	M230	Bending pressure signal		
232	M231	Bending return signal		
233	M232	Bending return in-position signal		
234	M233	Bending start signal		
235	M234			
236	M235			
237	M236			
238	M237			
239	M238			
240	M239			
241	M240	Clear stacking 0 counter GI90	input	
242	M241	Clear stacking 1 counter GI91	input	
243	M242	Clear stacking 2 counter GI92	input	
244	M243	Clear stacking 3 counter GI93	input	
245	M244	Clear stacking 4 counter GI94	input	
246	M245	Clear stacking 5 counter GI95	input	
247	M246	Clear stacking 6 counter GI96	input	
248	M247	Clear stacking 7 counter GI97	input	
249	M248	Clear stacking 8 counter GI98	input	
250	M249	Clear stacking 9 counter GI99	input	

251	M250	Visual data, get the camera data processing signal	state	
252	M251	Visual data, get the camera data processing signal	state	
253	M252	Visual data, get the camera data processing signal	state	
254	M253	Visual data, get the camera data processing signal	state	
255	M254	Visual data, get the camera data processing signal	state	
256	M255	Visual data, get the camera data processing signal	state	
257	M256	Visual data, get the camera data processing signal	state	
258	M257	Visual data, get the camera data processing signal	state	
259	M258	Visual data, get the camera data Processing signal	state	
260	M259	Visual data, get the camera data processing signal	state	
261	M260	The current buffer state put object--1, not put--0	state	
262	M261	Current buffer state	state	
263	M262	Current buffer state	state	
264	M263	Current buffer state	state	
265	M264	Change buffer signal; Current buffer status	state	
266	M265	Buffer layer 1;current buffer state	state	

267	M266	Buffer layer 2 ;current buffer state	state	
268	M267	Buffer layer 3; current buffer state	state	
269	M268	Buffer layer 4 ;current buffer state	state	
270	M269	Buffer layer 5; current buffer state	state	
271	M270	IO access to corresponding number 0	input	
272	M271	IO access to corresponding number 1	input	
273	M272	IO access to corresponding number 2	input	
274	M273	IO access to corresponding number 3	input	
275	M274	IO access to corresponding number 4	input	
276	M275	IO access to corresponding number 5	input	
277	M276	IO access to corresponding number 6	input	
278	M277	IO access to corresponding number 7	input	
279	M278	IO access to corresponding number 8	input	
280	M279	IO access to corresponding number 9	input	
281	M280	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
282	M281	Customer defined	input	Whether to stop the robot from editing PLC by itself

		alarm		
283	M282	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
284	M283	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
285	M284	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
286	M285	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
287	M286	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
288	M287	Customer defined alarm	input	Whether to stop the robot from editing PLC by itself
289	M288			
290	M289			
291	M290	Axis interference of J1 axis	state	
292	M291	Axis interference of J2 axis	state	
293	M292	Axis interference of J3 axis	state	
294	M293	Axis interference of J4 axis	state	
295	M294	Axis interference of J5 axis	state	
296	M295	Axis interference of J6 axis	state	
297	M296	Axis interference of J7 axis	state	
298	M297	Axis interference of J8 axis	state	
299	M298		state	
300	M299		state	
301	M300	Valid when reaching GP170	state	GP170 is valid after arrival, J1-J8 are all reached
302	M301	Valid when reaching GP171	state	GP171 is valid after arrival, J1-J8 are all reached

303	M302	Valid when reaching GP172	state	GP172 is valid after arrival, J1-J8 are all reached
304	M303	Valid when reaching GP173	state	GP173 is valid after arrival, J1-J8 are all reached
305	M304	Valid when reaching GP174	state	GP174 is valid after arrival, J1-J8 are all reached
306	M305	Valid when reaching GP175	state	GP175 is valid after arrival, J1-J8 are all reached
307	M306	Valid when reaching GP176	state	GP176 is valid after arrival, J1-J8 are all reached
308	M307	Valid when reaching GP177	state	GP177 is valid after arrival, J1-J8 are all reached
309	M308	Valid when reaching GP178	state	GP178 is valid after arrival, J1-J8 are all reached
310	M309	Valid when reaching GP179	state	GP179 is valid after arrival, J1-J8 are all reached
311	M310	Space interference	GP110 GP111	The robot enters the area GP110, GP111 is valid
312	M311	Space interference	GP112 GP113	The robot enters the area GP112, GP113 is valid
313	M312	Space interference	GP114 GP115	The robot enters the area GP114, GP115 is valid
314	M313	Space interference	GP116 GP117	The robot enters the area GP116, GP117 is valid
315	M314	Space interference	GP118 GP119	The robot enters the area GP118, GP119 is valid
316	M315	Space interference	GP120 GP121	The robot enters the area GP120, GP121 is valid
317	M316	Space interference	GP122 GP123	The robot enters the area GP122, GP123 is valid
318	M317	Space interference	GP124 GP125	The robot enters the area GP124, GP125 is valid
319	M318	Space interference	GP126 GP127	The robot enters the area GP126, GP127 is valid
320	M319	Space interference	GP128 GP129	The robot enters the area GP128, GP129 is valid
321	M320	Reach safety zone 0	state	After valid, interrupt the current process and call the safe0 program
322	M321	Reach safety zone 1	state	After valid, interrupt the current process and call the safe1

				program
323	M322	Reach safety zone 2	state	After valid, interrupt the current process and call the safe2 program
324	M323	Reach safety zone 3	state	After valid, interrupt the current process and call the safe3 program
325	M324	Reach safety zone 4	state	After valid, interrupt the current process and call the safe4 program
326	M325	Reach safety zone 5	state	After valid, interrupt the current process and call the safe5 program
327	M326	Reach safety zone 6	state	After valid, interrupt the current process and call the safe6 program
328	M327	Reach safety zone 7	State	After valid, interrupt the current process and call the safe7 program
329	M328	Reach safety zone 8	State	After valid, interrupt the current process and call the safe8 program
330	M329	Reach safety zone 9	State	After valid, interrupt the current process and call the safe9 program
331	M330	Station 1 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
332	M331	Station 2 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
333	M332	Station 3 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
334	M333	Station 4 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
335	M334	Station 5 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
336	M335	Station 6 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
337	M336	Station 7 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
338	M337	Station 8 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted

339	M338	Station 9 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
340	M339	Station 10 reservation running permission signal	input	Check before the reserved program starts running, and not check in the rest time; 1-permitted; 0-not permitted
341	M340	Reservation 0 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
342	M341	Reservation 1 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
343	M342	Reservation 2 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
344	M343	Reservation 3 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
345	M344	Reservation 4 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
346	M345	Reservation 5 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
347	M346	Reservation 6 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
348	M347	Reservation 7 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
349	M348	Reservation 8 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
350	M349	Reservation 9 station status	input	1--Normal; 0--Abnormal, not allowed to reserve
351	M350	Start button for reservation 1, input	input	M35X valid conditions: rising edge, keep the set valid time signal valid (operating parameter -20#), falling edge. When the station has no reservation, increase reservation. When the station is paused, continue to run. When the station reservation is waiting, cancel the reservation.
352	M351	Start button for reservation 2, input	input	
353	M352	Start button for reservation 3, input	input	
354	M353	Start button for reservation 4, input	input	
355	M354	Start button for reservation 5, input	input	
356	M355	Start button for reservation 6, input	input	
357	M356	Start button for reservation 7, input	input	

358	M357	Start button for reservation 8, input	input	
359	M358	Start button for reservation 9, input	input	
360	M359	Start button for reservation 10, input	input	
361	M360	Start light for reservation 1, input	output	Operation parameter -30#, set the output mode of M36X: 0—On when run, off when stop; 1--The program is completed and output for 2 seconds; 2—On when in reservation, off when reservation is finished; No reservation:0-off, 1-off, 2-off; Reservation in waiting: 0-flashing,1-off, 2-off; Reservation in running: 0-on,1-off, 2-on; Reservation in stopping: 0-off, 1-on when finish running, after 2s, it will be off, 2-on.
362	M361	Start light for reservation 2, input	output	
363	M362	Start light for reservation 3, input	output	
364	M363	Start light for reservation 4, input	output	
365	M364	Start light for reservation 5, input	output	
366	M365	Start light for reservation 6, input	output	
367	M366	Start light for reservation 7, input	output	
368	M367	Start light for reservation 8, input	output	
369	M368	Start light for reservation 9, input	output	
370	M369	Start light for reservation 10, input	output	
371	M370	Stop button of reservation 1, input	input	When the corresponding station is running, stop running. No action in other states
372	M371	Stop button of reservation 2, input	input	
373	M372	Stop button of reservation 3, input	input	
374	M373	Stop button of reservation 4, input	input	
375	M374	Stop button of reservation 5, input	input	
376	M375	Stop button of reservation 6, input	input	
377	M376	Stop button of	input	

		reservation 7, input		
378	M377	Stop button of reservation 8, input	input	
379	M378	Stop button of reservation 9, input	input	
380	M379	Stop button of reservation 10, input	input	
381				
382	M380	Stop light of reservation 1, output	output	When the reservation is stopped, the stop light is on
383	M381	Stop light of reservation 2, output	output	
384	M382	Stop light of reservation 3, output	output	
385	M383	Stop light of reservation 4, output	output	
386	M384	Stop light of reservation 5, output	output	
387	M385	Stop light of reservation 6, output	output	
388	M386	Stop light of reservation 7, output	output	
389	M387	Stop light of reservation 8, output	output	
390	M388	Stop light of reservation 9, output	output	
391	M389	Stop light of reservation 10, output	output	
392	M390	Cancel all reservations in current reservation waiting	input	
393				
394	M400	Current buffer status	state	Put object--1; not put--0
395	M401	Current buffer status	state	Put object--1; not put--0
396	M402	Current buffer status	state	Put object--1; not put--0

397	M403	Current buffer status	state	Put object--1; not put--0
398	M404	Current buffer status	state	Put object--1; not put--0
399	M405	Current buffer status	state	Put object--1; not put--0
400	M406	Current buffer status	state	Put object--1; not put--0
401	M407	Current buffer status	state	Put object--1; not put--0
402	M408	Current buffer status	state	Put object--1; not put--0
403	M409	Current buffer status	state	Put object--1; not put--0
404	M410	Current buffer status	state	Put object--1; not put--0
405	M411	Current buffer status	state	Put object--1; not put--0
406	M412	Current buffer status	state	Put object--1; not put--0
407	M413	Current buffer status	state	Put object--1; not put--0
408	M414	Current buffer status	state	Put object--1; not put--0
409	M415	Current buffer status	state	Put object--1; not put--0
410	M416	Current buffer status	state	Put object--1; not put--0
411	M417	Current buffer status	state	Put object--1; not put--0
412	M418	Current buffer status	state	Put object--1; not put--0
413	M419	Current buffer status	state	Put object--1; not put--0
414	M420	Customer user-defined alarm	input	
415	M421	Customer user-defined alarm	input	
416	M422	Customer user-defined alarm	input	
417	M423	Customer user-defined alarm	input	
418	M424	Customer user-defined alarm	input	
419	M425	Customer user-defined alarm	input	

420	M426	Customer user-defined alarm	input	
421	M427	Customer user-defined alarm	input	
422	M428	Customer user-defined alarm	input	
423	M429	Customer user-defined alarm	input	Not enabled, brake detection (M16) is valid! Please connect the brake power (BK-POWER) or check the brake board and wiring
424				
425	M440	GP170 ;J1-J6 position reached	state	
426	M441	GP171 ;J1-J6 position reached	state	
427	M442	GP172 ;J1-J6 position reached	state	
428	M443	GP173 ;J1-J6 position reached	state	
429	M444	GP174;J1-J6 position reached	state	
430	M445	GP175;J1-J6 position reached	state	
431	M446	GP176;J1-J6 position reached	state	
432	M447	GP177;J1-J6 position reached	state	
433	M448	GP178 ;J1-J6 position reached	state	
434	M449	GP179 ;J1-J6 position reached	state	
435	M450	GP96 ;J1-J6 position reached	state	
436	M451	GP97 ;J1-J6 position reached	state	
437	M452	GP98 ;J1-J6 position reached	state	
438	M453	GP99 ;J1-J6 position reached	state	
439	M454	Valid, alarm when J7	input	

		axis move		
440	M455	Valid, alarm when J8 axis move	input	
441				
442	M456	Arc tart output	output	
443	M457	Swing arc output starting arc	output	
444	M458	High-speed output	output	
445	M459			
446	M460	Output during running	output	
447	M461	Buzzer output	output	
448	M462	Circular arc output		
449	M463			
450	M464	Select posture output	input	
451	M465	Select joystick output	input	
452	M466	Invalid XYZ, valid ABC, select input	input	
453	M467			
454				
455	M481	A1sheet V1sheet	input	Valid, call corresponding file
456	M482	A2sheet V2sheet	input	Valid, call corresponding file
457	M483	A3sheet V3sheet	input	Valid, call corresponding file
458	M484	A4sheet V4sheet	input	Valid, call corresponding file
459	M485	A5sheet V5sheet	input	Valid, call corresponding file
460	M486	A6sheet V6sheet	input	Valid, call corresponding file
461	M487	A7sheet V7sheet	input	Valid, call corresponding file

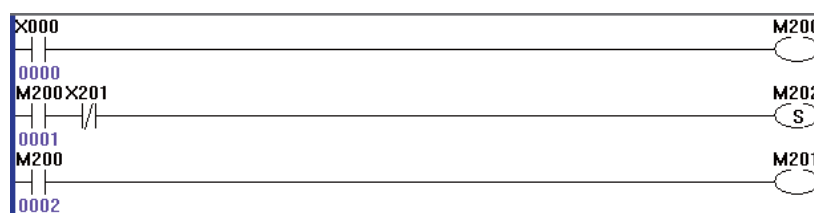
462	M488	A8sheet V8sheet	input	Valid, call corresponding file
463	M489	A9sheet V9sheet	input	Valid, call corresponding file
464	M490	A10sheet V10sheet	input	Valid, call corresponding file
465	M491	A11sheet V11sheet	input	Valid, call corresponding file
466	M492	A12sheet V12sheet	input	Valid, call corresponding file
467	M493	A13sheet V13sheet	input	Valid, call corresponding file
468	M494	A14sheet V14sheet	input	Valid, call corresponding file
469	M495	A15sheet V15sheet	input	Valid, call corresponding file
470	M496	A16sheet V16sheet	input	Valid, call corresponding file

6. Editing of PLC program

PLC trapezoidal editing of the CD60-CRX8 control system can be transmitted to the system for running through using the special PLC editing software (PLC.EXE) to edit and produce PLC files plc.lad (ladder diagram file) and plc.plc (command list file), or directly edit the ladder diagram to run on the system.

6.1 PLC ladder diagram examples

A. Example of reading rising edge



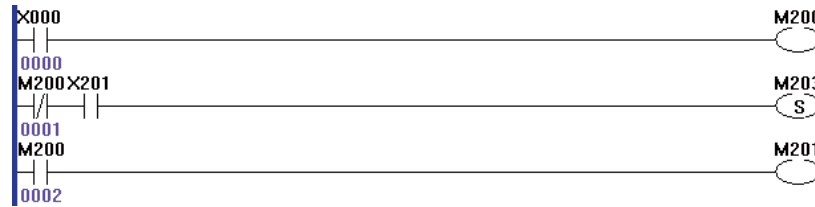
Work description: Use PLC to execute sequentially within a scan cycle to read rising or falling edges. X00 is the input port that needs to be read, and M200 and M201 are auxiliary relays. The rising edge of the read is stored in M202. When it needs to be read again, M202 needs to be reset (program or PLC reset).

When the first line is executed in sequence, if X00 changes from invalid to valid, M200 is valid.

In the second line, as the previous M200 is already valid, the normally open M200 in the second line is turned on, and the normally closed M201 has not been executed yet, and the normally closed M201 is in the on state. At this time, the M202 is set to be valid.

When running to the third line, as the previous M200 is valid and the normally open M200 is valid, then M201 is valid.

B. Example of reading falling edge



Work description: Use PLC to execute sequentially within a scan cycle to read rising or falling edges. X00 is the input port that needs to be read, and M200 and M201 are auxiliary relays. The rising edge of the reading is stored in M203. When it needs to be read again, M203 needs to be reset (program or PLC reset).

When the first line is executed in sequence, if X00 changes from valid to invalid, M200 will change from valid to invalid.

In the second line, since the previous M200 is invalid, the normally closed M200 in the second line is connected. The previous cycle M201 has been valid, and the current period has not been executed yet, and the normally open M201 is in the connected state. At this time, set M203 in valid.

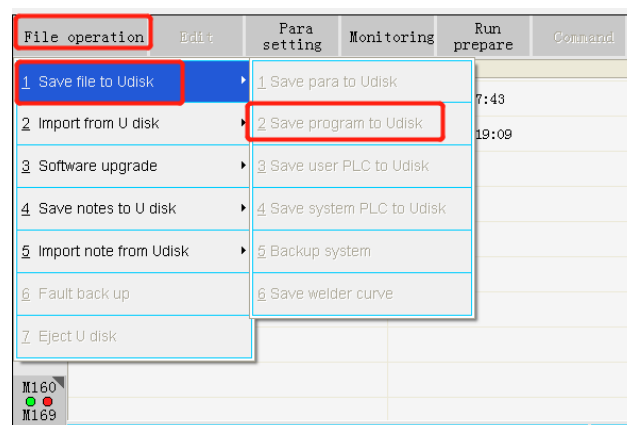
When running to the third line, because M200 was invalid before, M201 is invalid.

7. PLC program backup and update

In order to easily editing the system PLC program on the computer, the system can backup the PLC program to the U disk through the U disk, and when the editing is completed, the PLC can also be updated through the U disk to achieve the logic function that the user wants.

7.1 PLC backup

1. Insert the U disk into the USB port of the system host
2. Click <File Operation>-<Save File to U Disk>-<Save Program to U Disk>, as shown in the figure below:

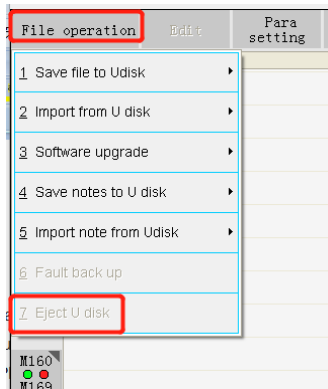


3. After a few seconds, the system (right below) will prompt "The plc.lad file has been copied to the E disk, if you want to unplug the U disk, please uninstall the U disk" and "plc.plc file has been copied to the E disk, if you want to

unplug Please uninstall the USB flash drive" as shown in the figure below:

 9 06-11 18:43:42 456 plc.plc文件夹已经拷贝到E盘下, 如要拔出请卸载U盘...

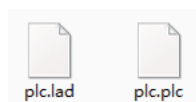
4. To uninstall the U disk, click <File Operation>-<Reject U Disk> in turn, as shown in the figure below:



System status bar prompt

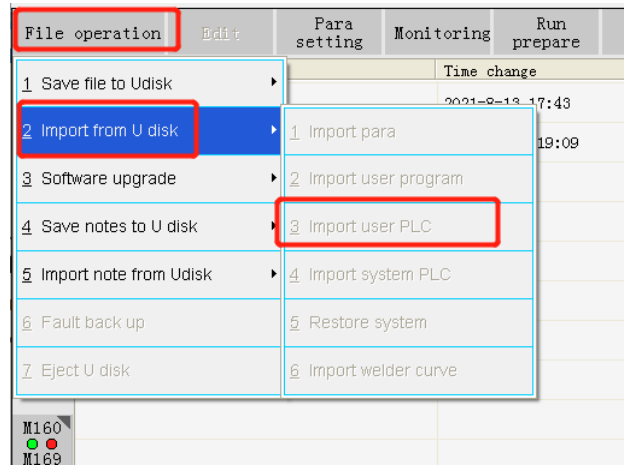
 10 06-11 18:52:53 445 U盘卸载成功

5. Check the backup files. Pull out the U disk from the system host, insert the computer, and open the U disk. There are files as shown in the figure below:



7.2 PLC update and replacement

1. Insert the U disk into the USB port of the system host
2. Click <File Operation>-<Import from U Disk>-<Import User PLC> in turn, as shown in the figure below:



3. After the user PLC reads into the system, the interface will show the restart confirmation remind.
4. Click <OK> to restart the system.
5. After waiting for the normal startup, the user PLC program update and replacement are completed, and the U disk is unplugged.